What changes an asteroid’s path?

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain how gravity can change velocity without contact.

SUCCESS CRITERIA - I can...

1. Draw velocity tangent to a path and force toward the attracting body.
2. Distinguish motion from a change in motion.
3. State what a straight-line model leaves out.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “gravity” and “acceleration” to explain your model or result. Sentence starters are optional.

gravity: an attractive interaction between masses

acceleration: rate of change of velocity

model: a drawing, explanation or equation for a physical situation

01 Warm-up: bring back an earlier idea

A simplified model predicts position (-30, 50) km with bounds ±3 km in x and ±5 km in y. Write a short briefing giving both intervals and two reasons it is not a real asteroid collision forecast.

Sentence starter: I predict … . My model supports this because … .

Marzano check - circle: ① ② ③

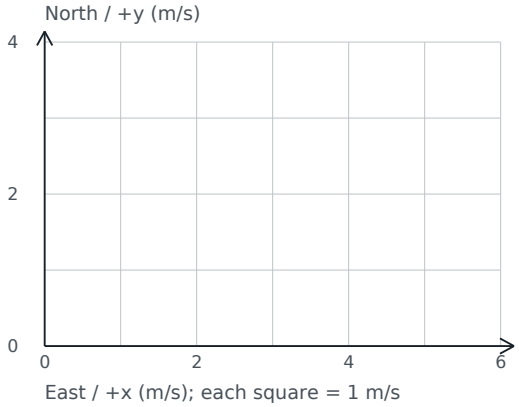
02 Worked example: explain the model

Reference: $a = F_{net}/m$.
Gravity attracts masses and can change speed, direction or both. Our asteroid scenario is fictional.

Worked reasoning: A fictional asteroid has a velocity arrow pointing right. A brief upward force gives it an upward change in velocity. The new velocity combines the original rightward motion with this change.

Your turn: A spacecraft is initially moving right. A short thruster burn adds an upward change in velocity.
Draw the original velocity and the change head-to-tail.
Draw the final velocity from the starting tail to the ending tip.

Sentence starter: The … in my model represents … . Its label shows … .



Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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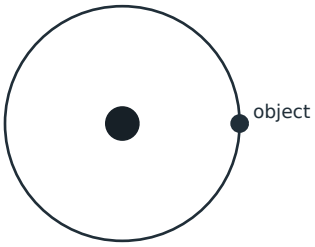
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Sketch the velocity and gravitational-force arrows at the marked point on the orbit. Why do they point in different directions?

Sentence starter: Both … . They differ in … because … .



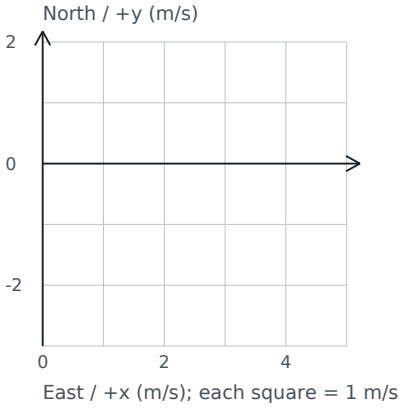
The center is the attracting body; the small dot on the circle is the orbiting object.

Marzano check - circle: ① ② ③

04 Apply the physics

A simulated probe starts with velocity (4, 0) m/s. Its thruster produces acceleration (0, -1) m/s² for 2 s. Plot the initial and final velocity vectors.
Find the final speed and explain whether speed stayed constant.

Sentence starter: Under the assumption …, the evidence … supports … because … .



Marzano check - circle: ① ② ③

05 Explain your reasoning

A student says space has no gravity because astronauts float.
Explain how orbiting astronauts can be falling.

Sentence starter: Under the assumption …, the evidence … supports … because … .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A 0.5 kg system accelerates at 0.7 m/s².
Predict acceleration for 0.9 kg at the same net force.
Explain the effect of greater resistance.
Use the table to keep the controlled and changed quantities distinct.

Word help: system = the object or objects chosen for analysis

Sentence starter: Both … . They differ in … because … .

Trial	Mass	Net force	Acceleration
A	____	____	____
B	____	____	____

Marzano check - circle: ① ② ③

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Continue down each column.

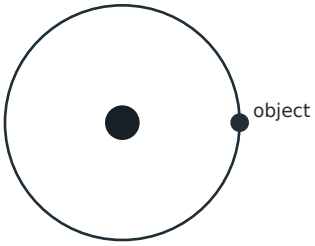
Sentence starters are optional. You may plan in any language.

07 Exit check

A force stays perpendicular to velocity during ideal circular motion.
Explain what changes and what stays constant.

Word help: ideal = using stated simplifying conditions

Sentence starter: Under the assumption …, the evidence … supports … because … .

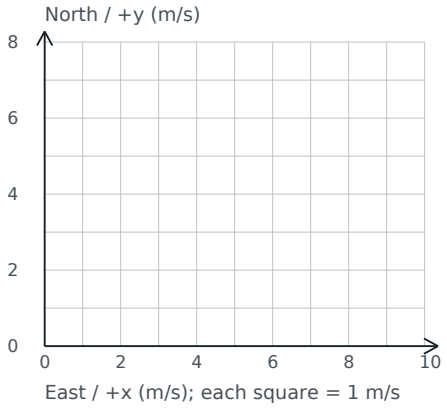


Marzano check - circle: 1 2 3

08 STRETCH: test the model

A simulated probe starts with velocity (8, 0) m/s. Its thruster produces acceleration (0, 2) m/s² for 3 s.
Draw the change in velocity head-to-tail with the initial velocity.
Find the final components and speed.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: 1 2 3